WORKSHOP GLOSSARY

ML in Cybersecurity

Can AI detect a suspicious login? The 14 terms you need today, in plain English.

THE BASICS

Machine learning (ML) – Software that learns patterns from examples instead of following hand-written rules.

Feature – One clue about a login, such as `failed_attempts`. *We use 6.*

Label – The “answer” for each example. *0 = normal, 1 = suspicious.*

TRAINING AND THE MODEL

Training data – The 80% the model learns from. The revision material. *800 logins.*

Test data – The 20% hidden during learning, used to check the model. The exam. *200 logins.*

Decision Tree – A chain of yes/no questions, such as “5 or more failed attempts?”. Easy to explain. *Our model.*

SCORING AND MISTAKES

Accuracy – Share of all predictions that were correct. Can hide what kind of mistakes were made. *Ours: 89.5%.*

Confusion matrix – A 2x2 table counting right and wrong predictions by type.

False positive – A normal login wrongly flagged. A false alarm. *Ours: 15.*

False negative – A suspicious login missed. The dangerous mistake. *Ours: 6.*

Precision – When the model flags a login, how often is it right? *Ours: 88%.*

Recall – Of all truly suspicious logins, how many did it catch? *Ours: 95%.*

Suspicious score – The model’s 0–100% output. A risk signal, not proof.

SECURITY

MFA – Multi-factor authentication: extra proof beyond a password, such as a phone code.